

Improving Reliability of State Estimation by Particle Filter with Cauchy Likelihood Function

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Abstract—Particle filtering, a sequential Monte Carlo method, is a powerful framework that the posterior probability density function (PDF) of the state can be approximated by a set of random particles and their associated weights. Particle filter (PF) is composed of these steps: particle initialization, prediction, updating, and resampling. For updating step, the framework needs to calculate the likelihood function which is based on the measurement model of system. When the measurement noise is considered additive and Gaussian, the conventional Gaussian error model is utilized. However, when the measurement noise level is low, Gaussian noise model provides large prediction error. We therefore aim to remedy the problem when the measurement noise level is low, instead of using standard likelihood calculation, we propose to implement Cauchy likelihood function (CLF) for particle weight calculation to resolve the problem. Via the root mean squared error (RMSE) and mean absolute error (MAE), simulation results show that it can provide comparatively better estimation performance than Gaussian likelihood function.

Index Terms—Cauchy likelihood function, Gaussian likelihood function, particle filter, state estimation, signal processing.

I. INTRODUCTION

It is advantageous in engineering and science fields that the prospective states of parameters can be calculated by analyzing their previous behaviors within a defined time frame. Yet, practically, the measurements obtained from dynamic systems are polluted with varying intensities of noise. Therefore, it is a challenging task to receive the state evolution accurately [1].

Particle filtering (PF) is a type of sequential Monte Carlo (SMC) method used to approximate the PDF of the parameters of interest in each time step after extracting the desired information from noisy data. In each step, PF computes the probability mass function (PMF) which is used to find PDF estimation using sequential importance sampling (SIS) algorithm, consisting a vector where the weights of each particle is recursively stored and weighted. It is known that the more the number of particles are, the better the system can estimate despite the high computational costs [2].

Particle filters are usually employed in the field of signal processing and estimation when the dynamic system states are expected to be estimated from the noisy measurements from a variety of sensors, which assists researchers in battery remaining useful life (RUL) predictions [3], object movement or action recognition [4], health care systems [5], etc [6].

There are four steps for typical implementation of particle filtering, namely, initialization, prediction, updating, resampling. Effective posterior PDF approximation can take the presence of a large number of particles to work with the

algorithm. In addition, lowering the number of particles used results in less probability of finding high-likelihood state values, which is the cause of particle degeneracy. That being the case, resampling is done after each step [7].

In updating the weight of particles, traditionally, Gaussian likelihood function (GLF) is applied for its consistency and compact variance for root mean squared errors and mean absolute errors [8]. When calculating the particle weight, in GLF, the weight is inversely proportional to the square of standard deviation (σ) in the Gaussian distribution. As a consequence, the smaller σ is, the larger the particle weight will be and the more probability that computer can face limitations as stated in [9].

In Cauchy distribution, the mean and mode are equal and expected value does not exist. Similar to Gaussian distribution, Cauchy distribution contains two parameters to formulate the PDF; a scale (s) parameter and a location parameter ($\mathbf{x}$), where Gaussian distribution requires standard deviation (σ) and a location parameter. When we compare the shape of both functions, we observe that Cauchy has thinner peak and wider tail than that of Gaussian [10] [11]. According to this observation, the use of CLF in the PF is expected to enhance the performance of the filter when the noise level in the observed signal is low. Therefore, we investigate in this paper the feasibility of CLF in PF that is handling the system under low noise level in the measurement. We also show in the paper that the use of CLF provides beneficial effect on tracking performance of the filter [12] [13].

The rest of the paper is structured as follows. In Section II, we discuss background theory and basic theorems of sequential Bayesian filtering. Then, in Section III, we provide the detail about CLF and its advantages and disadvantages over GLF. Simulation results and performance comparison of PF when using CLF and GLF will be discussed in Section IV. Finally, Section V concludes the paper.

II. SEQUENTIAL BAYESIAN FILTERING

Sequential Bayesian filtering is formulated by dynamical system by using state-space equations to interpret the evolution of the state vector, and to model the relationship between the state and measured data, respectively. The first equation is called state-space transition equation that describes how the state vector is updated from the previous time step. The second equation is called measurement (observed data)

equation as described earlier. The two equations are as follows:

$$\mathbf{x}_k = \mathbf{f}_{k-1}(\mathbf{x}_{k-1}, \mathbf{m}_{k-1}), \quad (1)$$

and

$$\mathbf{y}_k = \mathbf{g}_k(\mathbf{x}_k, \mathbf{n}_k). \quad (2)$$

Vectors $\mathbf{x}$ and $\mathbf{y}$ refer respectively to a state vector containing hidden states and observed data of systems that arrive sequentially in each time step, denoted by k . The notations $\mathbf{m}_k$ and $\mathbf{n}_k$ are noise quantities that are involved to state and observation equations in Eqs: 1 and 2, respectively. The functions $\mathbf{f}_{k-1}(\cdot)$ and $\mathbf{g}_k(\cdot)$ refer to state equation and measurement equation, which are assumed to be known multivariate functions with nonlinear properties.

The quantity N_p stands for the number of particles used, w_k^i refers to the normalized weight of the i^{th} particle at timestep k , and δ is the Dirac delta function. According to Bayes' theorem, posterior PDF of the state can be written as

$$p(\mathbf{x}_k | \mathbf{y}_k) \approx \sum_{i=1}^{N_p} w_k^i \delta(\mathbf{x}_k - \mathbf{x}_k^i). \quad (3)$$

In each round, since Bayesian filtering uses sequential importance sampling (SIS), the outcome of previous particle's weight is recursively used to find the next one. Equation (3) can be updated using the following from to find the weight of each particle as:

$$w_k^i \propto \frac{p(\mathbf{x}_k^i | \mathbf{y}_k)}{q(\mathbf{x}_k^i | \mathbf{y}_k)}. \quad (4)$$

After each loop, the particle weights should be updated by multiplying the posterior PDF with the weight from the previous time step $k-1$ in this case:

$$w_k^i \propto w_{k-1}^i \times p(\mathbf{y}_k | \mathbf{x}_k^i) \quad (5)$$

which, practically, can be reduced to

$$w_k^i \propto p(\mathbf{y}_k | \mathbf{x}_k^i). \quad (6)$$

If Gaussian likelihood function is used to find the PDF, the weight of each particle will be found by using the standard deviation of the value of GLF [14].

$$w_k^i = \frac{1}{\sqrt{2\pi}\sigma} \times \exp \left\{ \frac{-(\mathbf{y}_k - \mathbf{g}_k(\mathbf{x}_k^i))^2}{2\sigma^2} \right\} \quad (7)$$

In the Eq. (7), to find the predicted measurement of particle we use $\mathbf{g}(\mathbf{x}_k)$ so that it transforms $\mathbf{x}_k$ into a form with the same unit as $\mathbf{y}_k$.

After some iterations, we need to perform resampling so that SIS would not work well by failing to recognize only the high-weight particles, ignoring other slightly light-weighted particles [15]. When this occurs, the posterior PDF fails and this failure is called particle degeneracy. To decide when resampling is needed, we need to calculate whether the number of effective particles falls below a certain threshold using:

$$N_{eff} = \frac{1}{\sum_{i=1}^N (w_k^i)^2} \quad (8)$$

However, traditional resampling destructs particle diversity and resampled particles can be less likely to discover high-likelihood state values. Therefore, in the resampling step, we adopt the scheme used in [2] as a replacement to traditional resampling to ensure particle diversity and higher probability to discover high-likelihood state values.

III. CAUCHY LIKELIHOOD FUNCTION

Cauchy distribution (also known as Lorentz distribution) resembles Gaussian distribution in a way that both these distributions are symmetric around the middle. If both of these distributions have the same peak as in Fig. 1, the cumulative probability are similar to some extent [16].

However, it differs from normal distribution by having a comparative larger area around the tails of distribution [17]. Particle dispersion occurs in both Gaussian and Cauchy, especially when the scale parameters and standard deviation are small, but CLF demonstrates a better relief than the dispersion occurred by using GLF. This is because with the higher slope of CLF, each particle weight changes accordingly with the state value, and on the other hand, since many of these particles have very small weight, CLF performs better [9].

Cauchy distribution is seen in communication signals where the ratio of two inputs give the recorded observation, but when these signals are absent, since the errors from two inputs have their own normal distributions with zero means, the noise level will represent a Cauchy distribution [18] [19].

To find the PDF of Cauchy distribution, the following particle weight is used instead of the weight in Eq. (7) as follows [20]:

$$w_k^i = \frac{1}{\pi} \left(\frac{s}{s^2 + (\mathbf{y}_k - \mathbf{g}_k(\mathbf{x}_k^i))^2} \right). \quad (9)$$

Here, s is the scale parameter which indicates the value of half the maximum height at the half total width of the PDF [21] [22]. According to [9], in CLF, the standard deviation value can be derived from traditional GLF as

$$s = \sqrt{\frac{2}{\pi}} \sigma, \quad (10)$$

where σ in Eq. (10) refers to the standard deviation value in GLF as in Eq. (7). We can get the same peak value when the location parameters are identical and s obtained by multiplying σ with $\sqrt{\frac{2}{\pi}}$ or 0.8.

Key Differences between CLF and GLF

The key differences between Cauchy and Gaussian distributions are that Cauchy distributions have heavier tail as displayed in Fig. 1, making it less prone to outliers. It may assign lower weights to extreme outliers, eliminating bias when eliminating parameters. However, the drawbacks

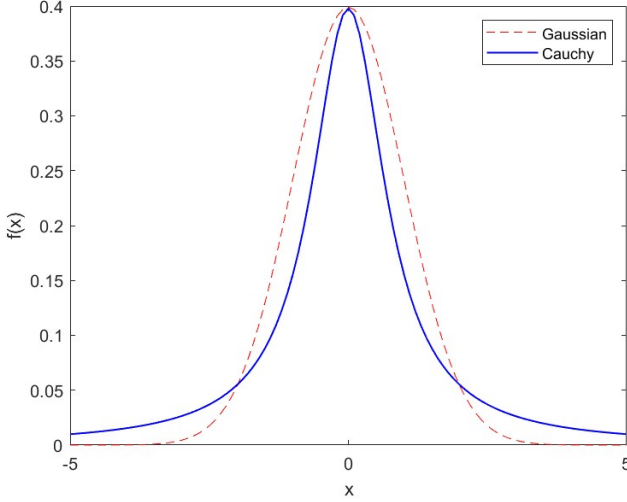


Fig. 1: Comparison of Gaussian and Cauchy distributions with equivalent standard deviation values.

are that it has less well-defined moments, which may be challenging for applications related to parameter estimation [23].

Gaussian distribution compared with Cauchy distribution can be defined as a thinner-tailed bell-shaped distribution, well-defined by their mean and variance. GLF may be stronger if maximum likelihood estimation (MLE) is used according to their distribution characteristics. Yet, the drawbacks are that they may be sensitive to extremities [24] [25].

IV. SIMULATION RESULTS

For simulation, we will be generating a synthetic signal with an abruptly changing values in a uni-dimensional state estimation [2] and perform state estimation by implementing both Gaussian and Cauchy likelihood functions in resampling step, and then comparing the final results as root mean squared errors (RMSE) and mean absolute errors (MAE).

- Firstly, we draw particles x_k^i as initial values for $i = 1, 2, 3, \dots, N_p$. The number of particles we will be using is 20000 in our simulation.
- Each particle is updated to become $x_k^i \sim p(x_k | x_{k-1}^i)$ using the univariate growth model [26]:

$$x_k = x_{k-1} + \frac{12x_{k-1}}{(1 + x_{k-1}^2)} + 7 \cos(1.2(k-1)) + m_k \quad (11)$$

and

$$y_k = 0.05x_k^2 + n_k. \quad (12)$$

The initial state x_0 is zero and m_k, n_k (the Gaussian, normally distributed noises added) are approximately equal to $\mathcal{N}(0, 2)$ and $\mathcal{N}(0, 1)$ respectively [27], [28].

- The weight of each particle (w_k^i) are found using Eqs. (6), (7), and (9) where k is a consistent time-step, σ_k^{state} as standard deviation using GLF and s_k^{state} as scale parameter using CLF. The sum of non-normalized weights for all time-steps are found and denoted as

w_k^{sum} . Also note that σ_{GLF} and s_{CLF} in this experiment will be considered small values. Because of abruptly jumping states in this example, the value of m_k is noticeably large.

- The effectiveness of each particle is measured using Eq. (8), where the ideal weight is the ones closest to $\frac{w_k^{sum}}{N_p}$. If the weight of a single particle is lower than the average of all particle weights, it is considered unsuitable or unfit for further calculations. These unfit particles are redrawn using

$$x_{G,k}^{new,i} \sim \begin{cases} \mathcal{U}[x_{G,k}^i - \sigma_{G,k}^{state}, x_{G,k}^{best} + \sigma_{G,k}^{state}] & \text{if } x_{G,k}^i < x_{G,k}^{best} \\ \mathcal{U}[x_{G,k}^{best} - \sigma_{G,k}^{state}, x_{G,k}^i + \sigma_{G,k}^{state}] & \text{if } x_{G,k}^i \geq x_{G,k}^{best} \end{cases} \quad (13)$$

for the particle weights found using GLF, and

$$x_{C,k}^{new,i} \sim \begin{cases} \mathcal{U}[x_{C,k}^i - s_{C,k}^{state}, x_{C,k}^{best} + s_{C,k}^{state}] & \text{if } x_{C,k}^i < x_{C,k}^{best} \\ \mathcal{U}[x_{C,k}^{best} - s_{C,k}^{state}, x_{C,k}^i + s_{C,k}^{state}] & \text{if } x_{C,k}^i \geq x_{C,k}^{best} \end{cases} \quad (14)$$

for those found using CLF. This procedure ensures that each particle weight is higher than $\frac{w_k^{sum}}{N_p}$.

- All particles weights can then be reset to $\frac{1}{N_p}$ and using the modified σ_k^{state} and s_k^{state} values, the particles are diversified with

$$x_{G,k}^i \sim \mathcal{N}(x_{G,k}^i, \frac{1}{(\sigma_{G,k}^{state})^2}) \quad (15)$$

and

$$x_{C,k}^i \sim \mathcal{N}(x_{C,k}^i, \frac{1}{(s_{C,k}^{state})^2}). \quad (16)$$

Eqs. (15) and (16) are for GLF and CLF respectively.

- For each time-step k , *maximum a posteriori* (MAP) estimation value for each x_k is found using the following expression [29]:

$$\hat{x}_{MAP} = \arg \max_{x_k^i} p(x_k | y_k). \quad (17)$$

- Finally, we move onto another time-step ($k + 1$) and recursed for another set of particles until k reaches its maximum (in this case, 100).

The results of comparison of MAP using CLF and GLF are shown in Figs. 2 and 3, respectively. The quantity σ used here is $\sqrt{0.1}$ or 0.3162. It is observable that in earlier time-steps, x_k estimation is not very different from one another, except that CLF catches up the abruptly changing state slower but in final 40, CLF shows comparatively improved results, with CLF barely being different from the true value.

For better comparison, we further observe the performance of the model via root mean squared error (RMSE) and mean absolute error (MAE) obtained from both likelihood functions. The RMSE is calculated as

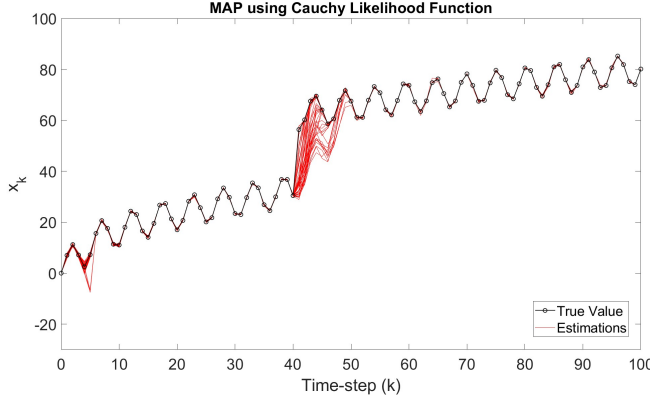


Fig. 2: MAP estimates using Cauchy Likelihood Function.

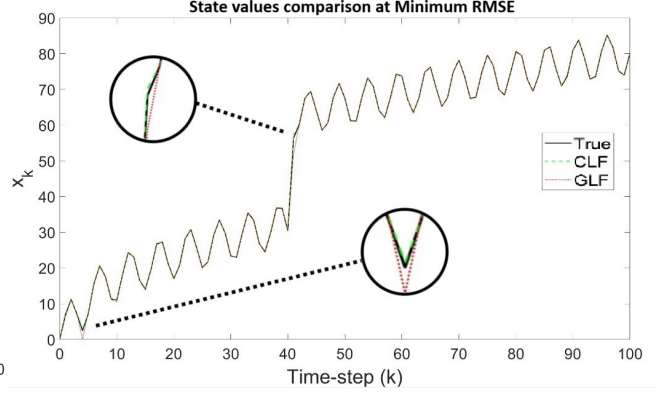
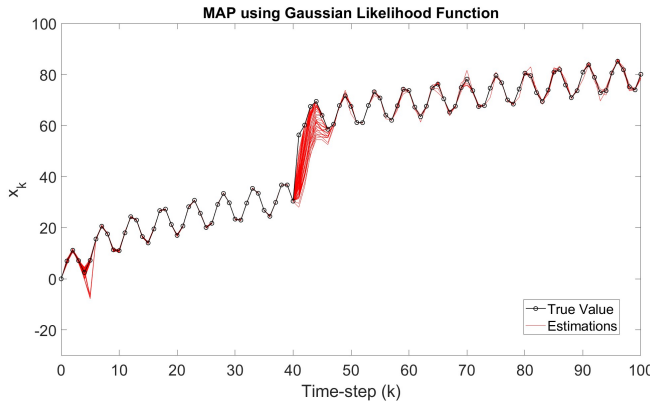
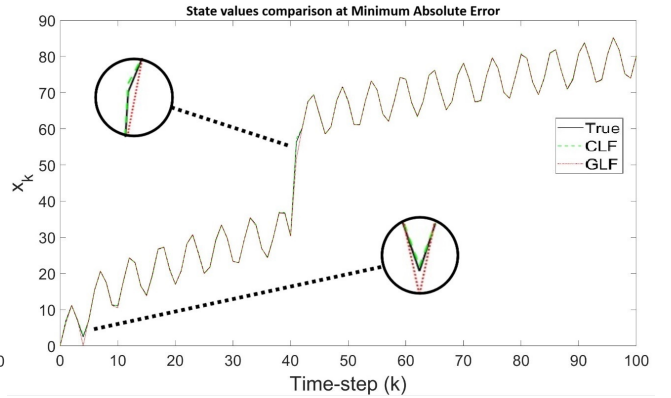

 Fig. 4: Comparison of state values ($\mathbf{x}_k$) for CLF and GLF at minimum RMSE.


Fig. 3: MAP estimates using Gaussian Likelihood Function.


 Fig. 5: Comparison of state values ($\mathbf{x}_k$) for CLF and GLF at minimum MAE.

$$RMSE = \frac{\sum_{r=1}^{r_{total}} \sqrt{\frac{\sum_{k=1}^{k_{total}} |x_{k,r} - \hat{x}_{k,r}|^2}{k_{total}}}}{r_{total}}, \quad (18)$$

and the MAE is computed by

$$MAE = \frac{\sum_{k=1}^{k_{total}} |x_{k,r} - \hat{x}_{k,r}|}{k_{total}}, \quad (19)$$

where $x_{k,r}$ is the estimated value and $\hat{x}_{k,r}$ is the true value.

In Figs. 4 and 5, we can see that in the run with the least RMSE and MAE, CLF can predict closer to the True value. When comparing the minimum RMSE and MAE for each standard deviation value, it is more obvious that at estimation using CLF can predict the state values much more accurately with lower error measures than using GLF, especially in abruptly jumping positions.

The comparison of average RMSE and MAE can be seen in Fig. 6(a) and (b) for different standard deviation values, ranging from $\sqrt{0.01}$ to $\sqrt{0.2}$ with 10000 particles for each time-step. When σ and s are low, the average RMSE and MAE of CLF is markedly lower and more stable than that of GLF. In Fig. 7(a) and Fig. 7(b), it indicates that comparison of

minimum RMSE and MAE indicates the difference between CLF and GLF values [30].

In Fig. 8, we compared minimum error measures with particle numbers 1000 to 25000 with σ_{GLF} value of $\sqrt{0.1}$. Though there are some fluctuations, it is still observable that CLF has lower minimum RMSE as shown in Fig. 8(a) and lower MAE as seen in Fig. 8(b) than those obtained from GLF. Therefore, these results prove that when low standard deviation likelihood values are present, CLF can give better performance when using an adaptive scheme, regardless of the number of particles used in the state estimation.

V. CONCLUSIONS

State estimation using particle filtering framework involves particle weight calculation. If the Gaussian environment is assumed, i.e., noise in the observed data is considered normally distributed, the Gaussian likelihood function is constructed. However, when performing state estimation under low measurement noise, we proposed using Cauchy distribution in this paper instead of Gaussian distribution in the likelihood formulation.

The simulation results show that the performance of the filter with CLF is superior than the filter with GLF under low measurement noise levels. Also, results indicate the consistency of the filter regardless of the number of particles since CLF delivers lower errors than those obtained from

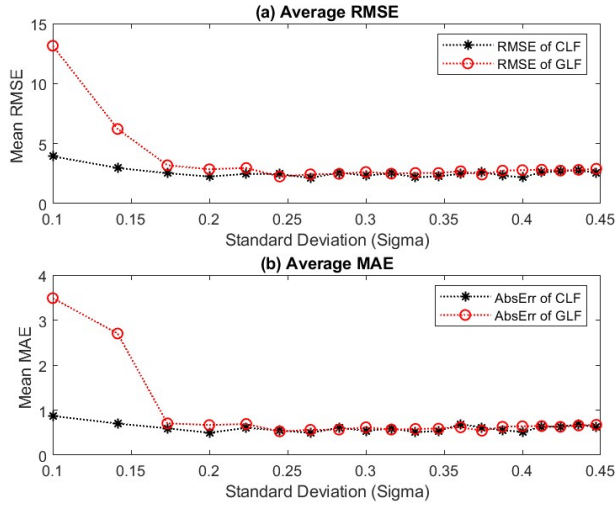


Fig. 6: Comparison of average: (a) RMSE and (b) MAE comparison of obtained from CLF and GLF for different standard deviation values.

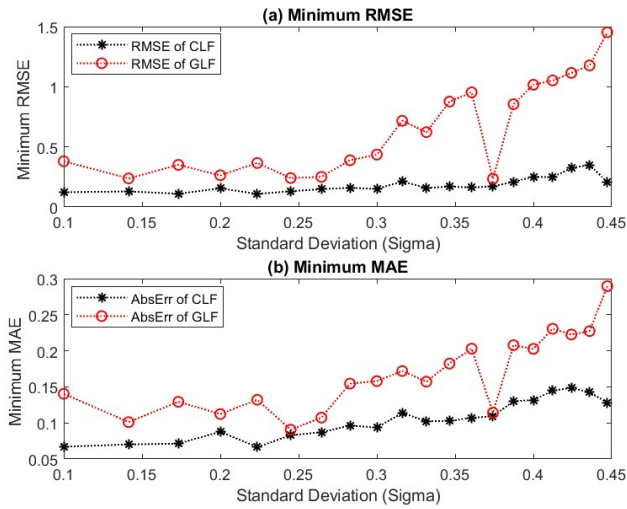


Fig. 7: Comparison of minimum: (a) RMSE and (b) MAE obtained from CLF and GLF for different standard deviation values.

GLF. Further, the CLF can be used for any cases other than abruptly changing states (results in this work) when small σ and scale parameter s are present in the measured signal. The extension of this work can be beneficial for many applications under this circumstance. For more implementation in various fields of particle filter usage, further studies should be made and this finding will be incorporated accordingly.

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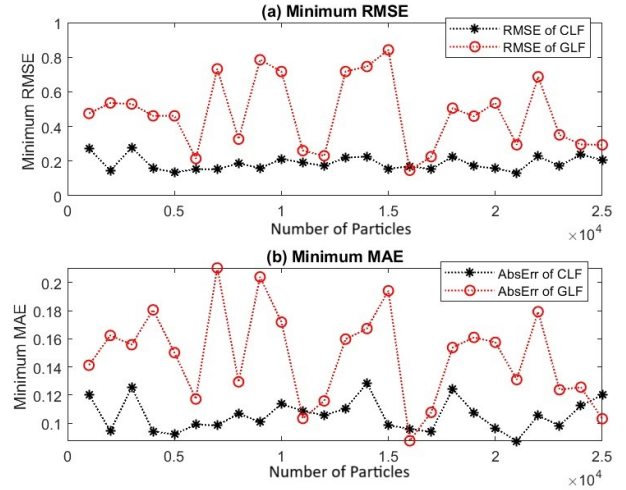


Fig. 8: Comparison of minimum: (a) RMSE and (b) MAE obtained from CLF and GLF for different number of particles.

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